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THIRD SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

INTRODUCTION TO AIML

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Outline the objectives of target function.	Understanding(2)	CO1	**02**
	b.	Describe the importance of feature selection	Remembering(1)	CO1	**02**
	c.	State about supervise and unsupervised learning.	Remembering(1)	CO2	**02**
	d.	Define ensemble learning.	Understanding(2)	CO3	**02**
	e.	List some merits of activation functions used in neural networks.	Remembering(1)	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Discuss about various components of Machine Learning (ML).	Understanding(2)	CO1	**05**
		OR			
	b.	Consider the dataset of exam scores for a class: 85, 90, 92, 78, 88. Calculate Mean, Median and Mode.	Applying (3)	CO1	**05**
3.	**a.**	Outline some Major Tasks in Data Preprocessing	Understanding(2)	CO2	**05**
		OR			
	b.	Discuss about Decision Tree with diagram.	Understanding(2)	CO2	**05**
4.	**a.**	Discuss on Hard and soft clustering.	Understanding(2)	CO3	**05**
		OR			
	b.	Find the distance between two points P (0, 4) and Q (6, 2) assuming any distance metric.	Applying (3)	CO3	**05**
5.	**a.**	Define the steps for finding Echelon form of matrix.	Remembering(1)	CO4	**05**
		OR			
	b.	Discuss on Eigen values and Eigen vectors of square matrices.	Understanding(2)	CO4	**05**
6.	**a.**	Summarize the working principle for RNN.	Understanding(2)	CO5	**05**
		OR			
	b.	Explain the activation function and its necessity.	Understanding(2)	CO5	**05**

7.	a.	Explain the Neural Network architecture with diagram.	Understanding(2)	CO5	**05**
		OR			
	b.	State about decision boundary and non-linear decision boundary.	Remembering(1)	CO5	**05**
		PART-C			**6 X 10=60**
8.	a.	Find linear regression equation for the following two sets of data:	Applying (3)	CO1	**10**

x	2	4	6	8
y	3	7	5	10

		OR			
	b.	Explain the Machine Learning process with a suitable diagram.	Understanding(2)	CO1	**10**
9.	a.	Explain the concept of Principal Component Analysis (PCA) with a diagram.	Understanding(2)	CO2	**10**
		OR			
	b.	Outline the comparisons between supervised, unsupervised, and reinforcement learning.	Understanding(2)	CO2	**10**
10.	a.	Discuss about Major Tasks in Data Preprocessing. What is the need of Feature selection?	Understanding(2)	CO2	**10**
		OR			
	b.	Explain the Decision Tree terminologies with suitable diagram and its importance in Machine Learning.	Understanding(2)	CO2	**10**
11.	a.	Cluster the following eight points (with (x, y) representing locations) into three clusters: A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9) Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2).The distance function between two points a = (x1, y1) and b = (x2, y2) is defined as- $P(a, b) = \|x2 - x1\| + \|y2 - y1\|$	Applying (3)	CO3	**10**
		OR			
	b.	Explain the clustering method with a suitable diagram. What is the importance of it?	Understanding(2)	CO3	**10**

12.	**a.**	Illustrate the steps to find the rank of a matrix by reducing it to the row echelon form.	Understanding(2)	CO4	**10**
		OR			
	b.	Write the Working procedure for problems using Gauss Seidel Method for example- 3X1+X2=9 X1+2X2=8	Applying (3)	CO4	**10**
13.	**a.**	Discuss the step-by-step flow of a basic neural network algorithm working process with diagram	Understanding(2)	CO5	**10**
		OR			
	b.	Identify and make a comparison list for CNN and RNN.	Understanding(2)	CO5	**10**